PUBLIC GOODS / WORKSHOP

Putting self-interest to work for the public good

An application of the Decentralized Curation framework

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Information becomes useful when it is fit for purpose.

Information quality depends on the user, the task and the context.

INTRINSIC

Accuracy · objectivity · credibility

CONTEXTUAL

Relevance · timeliness · completeness

REPRESENTATIONAL

Interpretability · consistency

ACCESSIBILITY

Availability · secure access

Accurate information can still be outdated, irrelevant or unusable.

Adapted from Wang & Strong (1996). Curation assesses these qualities against the requirements of the intended use.

Information quality is multidimensional.

ACCURACY
Are claims supported by evidence?

RELEVANCE
Do they bear on this decision?

TIMELINESS
Is the information current enough?

COMPLETENESS
Is decision-relevant context missing?

INTERPRETABILITY
Can the information be understood?

ACCESSIBILITY
Can intended users obtain and use it?

The purpose determines the quality profile. Assess each dimension separately.

The funding example separates accuracy from relevance; periodic reassessment keeps relevance current.

Curation creates value.
It also requires trust.

Delegating judgment makes us rely on the curator's:

HONESTY

Sincere reporting of their assessment.

COMPETENCE

Ability to understand and evaluate the evidence.

SCRUTINY

Critical examination of claims and evidence.

A capable curator still needs to investigate and report honestly.

These are distinct requirements. The mechanism should make all three worthwhile.

Who curates?
Under which incentives?

A CENTRALIZED CURATOR

Expertise, context and consistent judgment.

Limited knowledge and capacity.
Concentrated discretion and blind spots.

SIMPLE VOTE-COUNTING

Broad participation and equal votes.

A tally does not show knowledge,
investigation or strength of evidence.

Can we open participation and make informed judgment worth doing?

Democratic institutions can combine participation with expertise. Open entry alone does not ensure representation.

Change the incentives.
Change the outcome.

Game theory asks what participants gain from each strategy.

REWARD VOLUME

Encourage more submissions.

REWARD AGREEMENT

Encourage coordination.

REWARD CORRECTIONS

Encourage investigation.

The game aims to make honesty, competence and scrutiny the most rewarding way to participate.

Outcomes also depend on information, beliefs, costs and expectations. Rewarding agreement alone does not establish accuracy.

What will everyone choose?

53	55	57	59	61	63	65
67	69	71	73	75	77	79
81	83	85	87	89	91	93
95	97	99	100	101	103	105
107	109	111	113	115	117	119
121	123	125	127	129	131	133
135	137	139	141	143	145	147

Choose the number you expect everyone else to choose.

Think privately. No discussion.
On three, say your number together.

A Schelling point aligns expectations without requiring a binding agreement. Communication can help; salience alone does not establish truth.

How will everyone play?

Honest · Competent · Lazy	Deceptive · Uninformed · Thorough
Honest · Uninformed · Lazy	Honest · Competent · Thorough
Deceptive · Competent · Lazy	Honest · Uninformed · Thorough
Deceptive · Uninformed · Lazy	Deceptive · Competent · Thorough

Choose the way of playing you expect everyone else to choose.

Think privately. No discussion.
On three, say your choice together.

Shared context makes honesty, competence and scrutiny salient. The aim is to reward acting on them.

Expectations can reinforce good judgment.

1. EXPECTATION

I expect others to judge **honestly, competently and thoroughly.**

2. COHERENCE

I want to earn rewards by being **coherent with their judgments.**

3. MY STRATEGY

So I judge **honestly, competently and thoroughly too.**

The design goal: make informed judgment the most rewarding route to coherence.

Coherence means agreement between judgments. The reward structure must make careful assessment worth its cost.

From a curation problem to a mechanism.

1 DEFINE THE PROBLEM

Which decision needs better information?

2 CHOOSE THE QUALITIES

What makes information useful here?

3 CHECK FALSIFIABILITY

Can false claims be feasibly refuted?

4 DESIGN BY DIMENSION

Different judgments need different mechanisms.

Decentralized Curation: make useful assessment and correction personally rewarding.

Use challenge-based accuracy when falsification is feasible and cheaper than exhaustive verification.

One open pool. Continuous funding.

POOL CREATOR Publishes the goal and criteria

FUNDERS
Supply a budget

→

PUBLIC-GOODS POOL

Policy + current weights

Allocates by rule

→

BENEFICIARIES
Receive their shares

↑ ASSESSMENT AND CORRECTION

NOMINATORS
Claims, evidence and deposits

JURORS
Relevance, appeals and reassessment

FACT-CHECKERS
Dispute claims; an external arbitrator rules

Assessment costs need funding; a beneficiary budget can arrive later. Jury and appeal rounds still occur.

Set the purpose.
Back claims with evidence.

How much does this nominee's completed and maintained work underpin how research is done, shared and kept, now?

0.00	0.25	0.50	0.75	1.00
not infrastructure	matters to one field	relied on in several fields	widely relied on	critical across research

ILLUSTRATIVE NOMINEE / NUMPY 2.0.0

"Released on 16 June 2024."

EVIDENCE

Published NumPy release announcement

The date is checkable.
How much the work matters is a separate judgment.

The creator defines the policy. The nomination backs factual claims; the jury assesses relevance.

One jury.
Different judgments.

6 seats · 3 jurors

Juror A Seats: 3 — 0.700



Juror B Seats: 2 — 0.400



Juror C Seats: 1 — 0.700



MEAN WEIGHTED BY SEATS

0.600

(3 × 0.70 + 2 × 0.40 + 1 × 0.70) ÷ 6 = 0.600

Seal privately → reveal

Deposits enter a draw for paid jury work; one juror may hold several seats

Curated shares.
Ready for any budget.

WEIGHTS → SHARES OF ONE BUDGET

Contribution	Weight	Share	Units/day
NumPy	0.600	40.0%	**40.00**
Archive	0.300	20.0%	**20.00**
Registry	0.600	40.0%	**40.00**



NUMPY'S SHARE

40.0%

Illustrative distribution: **100** units/day

100 × 40.0% = 40.00 units/day

INITIAL ASSESSMENT; ELIGIBLE FOR FUNDING

Weights express relative priority, not measured public benefit. Set the budget to zero: the weights and shares remain.

Dispute a score?
Fund an appeal.

INITIAL JURY

6

seats · score **0.600**

→

CONTESTED APPEAL

12

seats · score **0.700**

8 at 0.800; 4 at 0.500

Competing fully funded scores go to a larger jury.

An appeal challenges an existing score; a later reassessment asks what matters now

Challenge claims. Reward fact-checking.

SUPPOSE A NOMINATION FALSELY CLAIMS

> “NumPy 2.0 was released in 2020.”
>
> The release record says 16 June 2024.

Challenge → External arbitrator →

Challenge succeeds?

Deposit to fact-checker; nomination ends

Successful fact-checker gets the deposit. Future funding stops; past earnings remain earned.

Reassess periodically.
Update the weights.

INITIAL SCORE

0.600

→

LATER SCORE

0.300

NUMPY / DAY

25.00

Lower weight takes effect immediately. Earlier accrual stays earned; only future accrual uses the new shares.

A fresh assessment updates current relevance; an appeal contests an existing score

Now you are the jury.



https://decentralized-curation.0xferit.workers.dev/r/leiden

Name → seat → score

No wallet. Workshop points only.

Equal voting weight every round, regardless of earlier wins or losses.

Judge the work.
Use the same question.

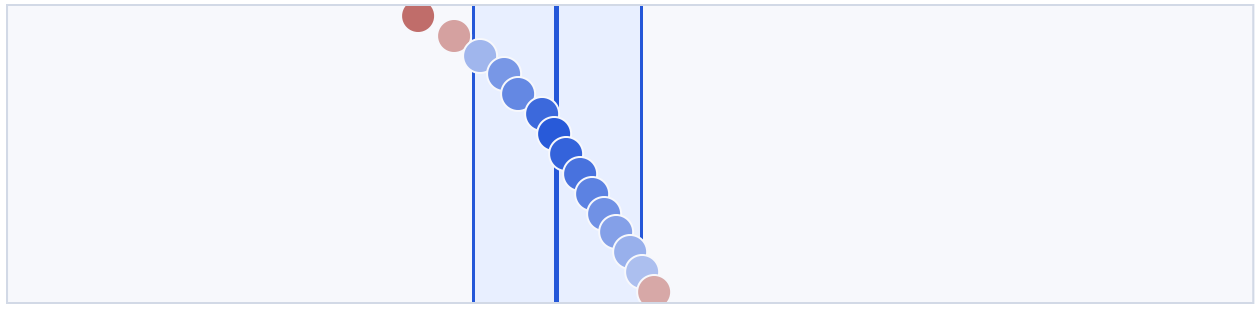
How much does this nominee’s completed and maintained work underpin how research is done, shared and kept, now?

0.00	0.25	0.50	0.75	1.00
not infrastructure	matters to one field	relied on in several fields	widely relied on	critical across research

how many rely · role in research · other options

Brief descriptions and prior knowledge; use the shared criteria and explain your evidence.

How your scores earn points.



0.00 Drag a juror; arrow keys also work 1.00

result	spread	zone	inside	losing points
0.44	0.055	±0.068	12	3

Blue: inside. Red: outside.

GROUP RESULT

Average of the scores

REWARD ZONE

Set by the spread of scores

POINTS

Closer to the result earns more

Points reward agreement; they do not establish truth, effort or expertise.

6 nominees.
Your judgment.

01
Inkscape

02
Linux

03
PDF

04
LaTeX

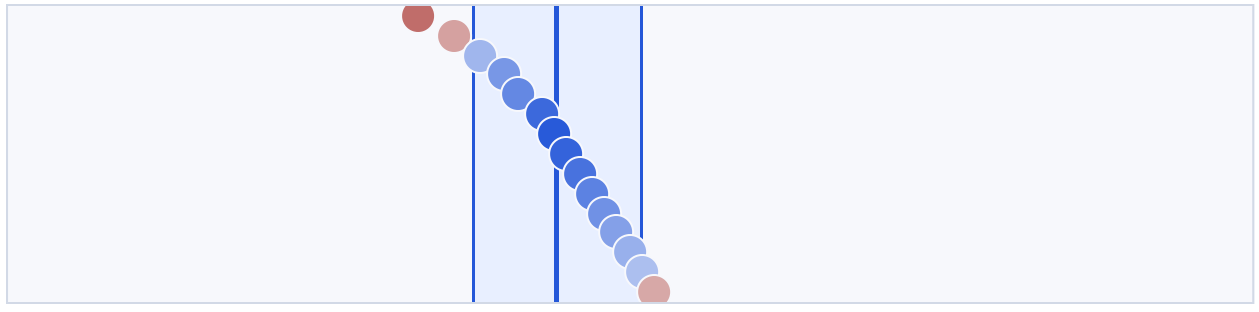
05
arXiv

06
NumPy

Read → Score alone → Seal → Open together

Keep the secret words used to seal your score private. Explain your evidence after revelation.

Did our scores follow the evidence?



0.00 Drag a juror; arrow keys also work 1.00

result	spread	zone	inside	losing points
0.44	0.055	±0.068	12	3

Blue: inside. Red: outside. Outline: assigned bloc at 0.90; lowest starting scores join first.

Bloc share of stake 0%

What did we reward?

What shaped the group result?

What would change your mind?

A well-informed minority can lose points. This exercise does not establish resistance to collusion.

Every defense has a boundary.

Attack	Response	Still exposed
Shift the criteria	Published goal and criteria	Bad policy; shared bias
Copy a score	Seal → reveal	Early reveals; coordination
Capture the jury	Deposit-based draw	Concentrated stake
Keep a bad ruling	Publicly funded appeal	Cost; shared bias
Prove a sealed score to a buyer	Workshop only: penalty + bounty	Other verification routes remain

Discuss which defenses help, and where each could fail.

Make scrutiny worthwhile.

Better information. Better decisions. Make personal benefit serve the public good.

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